SADITHA UDAYANGA

PYTHON & AI ENGINEER

AI Engineering • LLMs • RAG • AI Agents • Voice AI • Backend Systems
Colombo, Sri Lanka | tmsaditha@gmail.com | 0717893455
LinkedIn: saditha-udayanga | GitHub: github.com/saditha93



PROFESSIONAL PROFILE

Python & AI Engineer with 7+ years of software engineering experience across AI, voice automation, intelligent search, enterprise applications, and e-commerce platforms. Strong focus on Python, LLM integrations, RAG, MCP, AI agent skills, MongoDB Atlas, Redis, Docker, AWS, Linux, and API-driven systems. Experienced in architecture input, code reviews, delivery planning, mentoring, production troubleshooting, and stakeholder collaboration.

CORE TECHNICAL STRENGTHS

AI & LLM Engineering: LLM integrations, RAG, MCP, AI agent skills, AI workflows, intelligent search, voice AI, embeddings, voice search
Python & Backend: Python, Django, Flask, REST APIs, JSON/XML, API integrations, WebSocket concepts
Data & Infrastructure: MongoDB Atlas, Redis, MySQL, Docker, AWS, Linux, Nginx, LEMP, GitHub, GitHub Actions
AI / Voice: Voice agents, speech workflows, wake-word detection, voice search, LLM-based conversation flows
Engineering: Architecture input, code review, mentoring, technical planning, production debugging, stakeholder communication
Supporting Stack: React.js, Angular, JavaScript, TypeScript, Node.js, Next.js, Magento 2, GraphQL, PHP, Laravel

PROFESSIONAL EXPERIENCE

Software Engineer / AI Engineer | NeoSolax (Pvt) Ltd

Nov 2023 – Present | Mount Lavinia, Sri Lanka

- Develop and maintain systems across AI, voice automation, intelligent search, backend, e-commerce, and enterprise products.
- Design and maintain AI-driven systems including Recomdo AI smart search and Belp AI voice assistant capabilities using LLMs, MongoDB Atlas, Redis, RAG, MCP, and skills-based agent configuration.
- Contribute to architecture decisions, code reviews, delivery planning, debugging, production issue resolution, and developer guidance.
- Work with Docker, AWS, Nginx, Linux, GitHub, and database environments to support production-grade systems.
- Collaborate with business stakeholders to translate requirements into scalable, maintainable, and commercially useful technical solutions.

Senior Software Engineer | Directorate of Information Technology, Sri Lanka Army

Jan 2022 – Nov 2023 | Sri Lanka

- Developed and maintained Laravel-based government and military-related enterprise web applications.
- Built backend modules, administration workflows, database-driven features, reporting interfaces, and production support improvements using PHP, Laravel, MySQL, and LEMP.
- Contributed to Officer Management, National Food Security Programme, and related enterprise applications.

NeoSolax (Pvt) Ltd — Career Progression

Jun 2019 – Jan 2022 | Mount Lavinia, Sri Lanka

- Senior Software Engineer — Jan 2021 – Jan 2022: delivered Magento 2 e-commerce implementations, React/PWA frontends, GraphQL integrations, custom modules, and frontend/backend enhancements.
- Software Engineer — Jan 2020 – Jan 2021: developed Magento 2 full-stack features, frontend components, backend modules, checkout improvements, and production fixes.
- Associate Software Engineer — Jun 2019 – Jan 2020: supported e-commerce development, maintenance, bug fixing, and deployment tasks.

FEATURED AI PROJECTS

RECOMDO AI — AI-Powered Intelligent Search Platform

Python • LLMs • MongoDB Atlas • Redis • TensorFlow • Voice Search

- AI-powered smart search engine for e-commerce platforms using Python services, LLM-based search intelligence, MongoDB Atlas, and Redis.
- Worked on TensorFlow-based model development for voice search and wake-word identification to support voice-driven product discovery.

BELP AI — AI Voice Agent Platform

LLMs • RAG • MCP • Agent Skills • Voice Conversation Workflows

- AI assistant voice conversation platform for phone-call style customer interactions.
- Developed and configured agent behavior using RAG knowledge, MCP/tool integrations, skills, and LLM-based conversation workflows.
- Worked across backend APIs, AI orchestration, data services, and configurable conversation flows to support customer assistance and service guidance.

SELECTED PROJECT PORTFOLIO

Officer Management System (OMMS) — Laravel 9, PHP, MySQL, LEMP
Enterprise officer management system with administrative workflows and database modules.
National Food Security Programme — Laravel 8, PHP, MySQL, LEMP
Management system for data collection, tracking, and reporting.
ADCP Mail Server — LDAP, Dovecot, Postfix, Roundcube, PHP, MySQL, LEMP
Enterprise mail server implementation and configuration.
E-commerce Platforms — Magento 2, React/PWA, GraphQL, PHP, MySQL, AWS, Docker, Nginx
Selected platforms: Buyabans, Nomin E-Shop, TechZone, ApeKade, J20 Sports, Chihuly, and Arienti.lk.
API Solution — WSO2 Connectors — JSON, XML, WSO2 ESB, Postman
Implemented connector and proxy scenarios with API testing.
Poker Game Web Application — MEAN Stack, MongoDB, Express.js, Angular, Swagger, Socket.io
API development and real-time functionality support for a multiplayer platform.

EDUCATION

- MSc in Enterprise Applications Development — Sri Lanka Institute of Information Technology (SLIIT) | 2021 – Present
- BSc (Hons) in Computer Science — University of Jaffna | 2015 – 2019

ENGINEERING & ACHIEVEMENTS

- Experienced in architecture input, code reviews, mentoring, technical planning, and production problem solving.
- Participated in IEEExtreme 12.0 Programming Competition, CodeExpertz 1.0, Code Craft Competition, and Yarl Hackathon.
- Representative of the Computer Society, Faculty of Science, University of Jaffna; active IEEE member.
- University of Jaffna rugby, chess, and basketball team member; SLUG 2016 and Inter-University Games 2017 participant.

LANGUAGES

English • Sinhala • Tamil

PROFESSIONAL REFERENCES

H. P. G. Deshani Meronika	Senior Project Manager, NeoSolax (Pvt) Ltd | 0710975956 | deshani.hansamali@neosolax.com
Sahan Thilakarathna	Tech Lead, NeoSolax (Pvt) Ltd | 0713477808 | sahan.thilakarathna@neosolax.com